

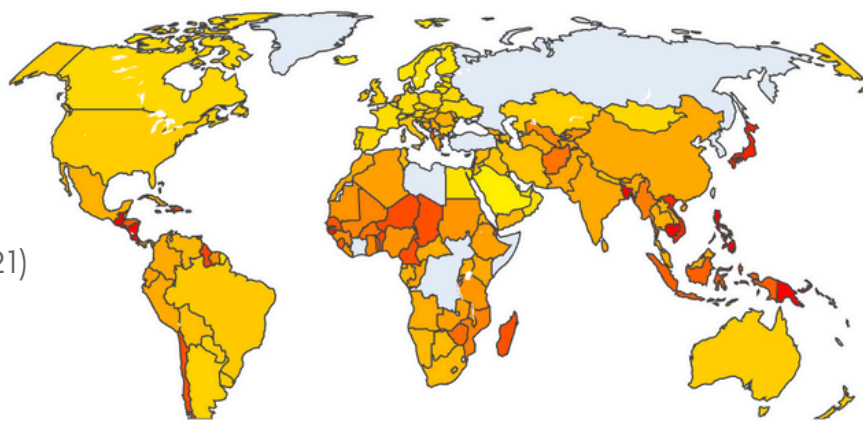
INFOVIS REDESIGN

Highlights

- Layered design to separate flood, heat, and drought risks
- Color scheme aligned with risk semantics
- Tooltip and interactive zoom for user-specific analysis
- Added metadata and source references

This project was guided by **several theories**:

- Tufte (1983)
- Munzner (2014)
- Fekete&Plaisant (1999)
- Buja et al. (1996)
- FAIR Principle
- Ziemkiewicz et al. (2021)



WRI

Tools Used

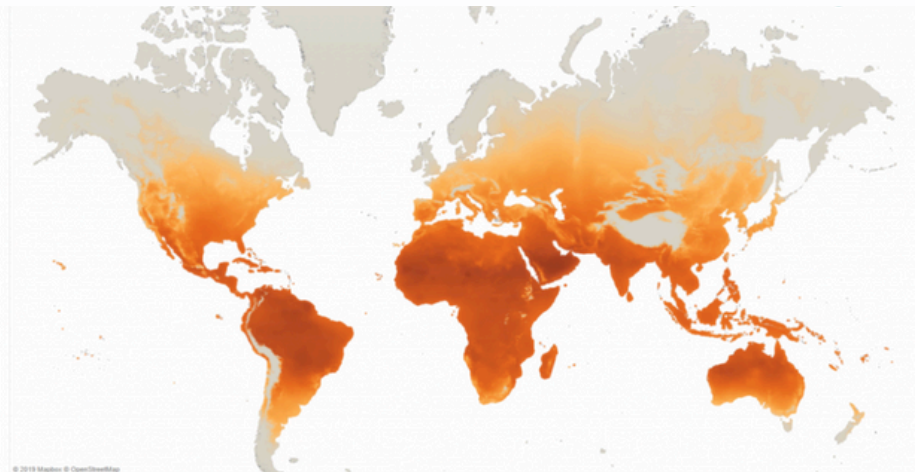
- Python: pandas, pycountry, Plotly



Redesigned: Interactive, layered visualization with clearer labeling and user guidance.

Workflow Pipeline

1. Data Processing: Cleaned and prepared CDD indicators using pandas
2. Country Mapping: Matched country names to ISO codes via pycountry
3. Visualization: Developed choropleth maps in Plotly for interactivity
4. User Options: Enabled dropdown filters (exposure, vulnerability, etc.)
5. Accessibility: Color-contrast checked; added tooltips and labels
6. Publication: Deployed code and dataset via GitHub for open access



Design Objectives

- Improve readability with clear labels and legends
- Apply emotional design principles to raise awareness
- Add interactivity for user-driven exploration
- Ensure metadata transparency (source, timeframe, indicators)
- Follow accessibility and perceptual optimization standards

Original: Lacked metadata and user interaction, difficult to interpret risk categories.